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Studierichting: Informatica
Oefeningenreeks 1D nr. 20.5

Herschrijven: $z^4 + 4 = z^4 + 4z^2 + 4 - 4z^2 = \left(z^2 + 2\right)^2 - (2z)^2$

$= \left(z^2 - 2z + 2\right) \cdot \left(z^2 + 2z + 2\right)$

discriminant en nulpunten:

$\left(z^2 - 2z + 2\right)$

$\Delta_1 = (-2)^2 - 4 \cdot 1 \cdot 2 = -4$

$\Rightarrow z_1 = \dfrac{-(-2) - \sqrt{-4}}{2 \cdot 1} = \dfrac{2 - 2i}{2} = 1 - i$

$\Rightarrow z_2 = \dfrac{-(-2) + \sqrt{-4}}{2 \cdot 1} = \dfrac{2 + 2i}{2} = 1 + i$

$\left(z^2 + 2z + 2\right)$

$\Delta_2 = 2^2 - 4 \cdot 1 \cdot 2 = -4$

$\Rightarrow z_3 = \dfrac{-2 - \sqrt{-4}}{2 \cdot 1} = \dfrac{-2 - 2i}{2} = -1 - i$

$\Rightarrow z_4 = \dfrac{-2 + \sqrt{-4}}{2 \cdot 1} = \dfrac{-2 + 2i}{2} = -1 + i$

References